Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: SB Farms

Physical address of dairy:

Number and Street	City	County	Zip Code
11450 Carroll RD	Elk Grove	Sacramento	95757

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 07/01/1979

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

X132-X240-XX43-XXXX

B. OPERATORS

Silva, James

Operator name: Silva, James

Telephone no.: Landline: ; Cellular: (209) 321-1318

Mailing Address Number and Street	City	State	Zip Code
22600 N Pearl RD	Acampo	CA	95220

This operator is responsible for paying permit fees.

Silva, Joe

Operator name: Silva, Joe

Telephone no.: Landline: ; Cellular: (209) 607-7197

Mailing Address Number and Street	City	State	Zip Code
12997 E Peltier RD	Acampo	CA	95220

C. OWNERS

Silva, James

Legal owner name: Silva, James

Telephone no.: Landline: | Cellular: (209) 321-1318

22600 N Pearl RD | Acampo | CA | 95220
Mailing Address Number and Street | City | State | Zip Code

This owner is responsible for paying permit fees.

Silva, Joe

Legal owner name: Silva, Joe

Telephone no.: Landline: | Cellular: (209) 607-7197

12997 E Peltier RD | Acampo | CA | 95220
Mailing Address Number and Street | City | State | Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	350	350	0	0
Number under roof	900	225	0	0	200	150
Maximum number	900	225	350	350	200	150
Average number	900	225	350	350	200	150
Avg live weight (lbs)	1,200	1,300	1,100	750		

Predominant milk cow breed: Holstein

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 34,596.87 *tons per reporting period*

Total nitrogen from manure: 427,219.59 *lbs per reporting period*

After ammonia losses (30% loss applied): 299,053.71 *lbs per reporting period*

Total phosphorus from manure: 69,027.03 *lbs per reporting period*

Total potassium from manure: 172,358.30 *lbs per reporting period*

Total salt from manure: 475,503.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,225,000 *gallons*

Total nitrogen generated: 38,790.66 *lbs*

Total phosphorus generated: 5,118.36 *lbs*

Total potassium generated: 35,243.02 *lbs*

Total salt generated: 185,018.66 *lbs*

	3,225,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	3,225,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well Field #3	Ground water
Ag Well Field #6	Ground water
Dairy/Dom. #1	Ground water
Dairy/Dom. #2	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/10/2023	Solid commercial fertilizer *Starter Fertilizer*	3.25 *ton*	Dry-weight	8.0	16.000000	38.000000	38.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	956.80	2,272.40	2,272.40	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	956.80	2,272.40	2,272.40	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/24/2023	Corral solids	2,000.00 *ton*	Dry-weight	22.4		13,000.00	3,510.00	20,200.00		0.00
10/11/2023	Corral solids	1,000.00 *ton*	Dry-weight	10.8		15,400.00	4,870.00	24,800.00		52.30

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	67,825.60	19,583.12	106,944.00	933,032.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	67,825.60	19,583.12	106,944.00	933,032.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #1	14	14	2	both	X132-0240-X020-XXXX
Field #10	18	18	2	both	X132-0240-X043-XXXX
Field #2	25	25	2	both	X132-0240-X020-XXXX
Field #3	42	42	2	both	X132-0240-X020-XXXX
Field #4	14	14	2	both	X132-0240-X020-XXXX
Field #5	25	25	2	both	X132-0240-X020-XXXX
Field #6	10	10	2	both	X132-0240-X020-XXXX
Field #7	40	40	2	both	X132-0240-X020-XXXX X132-0240-X034-XXXX
Field #8	25	25	2	both	X132-0240-X034-XXXX
Field #9	38	38	2	both	X132-0240-X034-XXXX
Totals for areas that were used for application	251	251	20		
Totals for areas that were not used for application					
Land application area totals	251	251	20		

B. CROPS AND HARVESTS

Field #1

Field name: Field #1

10/24/2022: Winter Forage

Crop: Winter Forage Acres planted: 14 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	224.00 *ton*	Dry-weight		50.4	21,600.00	2,260.00	19,400.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	342.40	35.20	307.20	0.00
Total actual harvest content	16.00	342.84	35.87	307.92	1,555.46

Field #1

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	392.00 *ton*	Dry-weight		70.6	13,100.00	3,110.00	18,000.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	215.60	50.40	280.00	0.00
Total actual harvest content	28.00	215.68	51.20	296.35	971.38

Field #10

Field name: Field #10

10/27/2022: Winter Forage

Crop: Winter Forage Acres planted: 18 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	288.00 *ton*	Dry-weight		51.8	11,100.00	2,420.00	17,400.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	171.20	36.80	265.60	0.00
Total actual harvest content	16.00	171.21	37.33	268.38	1,465.28

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	504.00 *ton*	Dry-weight		70.6	11,900.00	2,800.00	17,700.00		3.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	196.00	44.80	291.20	0.00
Total actual harvest content	28.00	195.92	46.10	291.41	609.17

Field #2

Field name: Field #2

10/24/2022: Winter Forage

Crop: Winter Forage Acres planted: 25 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	300.00 *ton*	Dry-weight		50.4	21,600.00	2,260.00	19,400.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	342.40	35.20	307.20	0.00
Total actual harvest content	12.00	257.13	26.90	230.94	1,166.59

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	700.00 *ton*	Dry-weight		70.6	13,100.00	3,110.00	18,000.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	215.60	50.40	280.00	0.00
Total actual harvest content	28.00	215.68	51.20	296.35	971.38

Field #3

Field name: Field #3

10/24/2022: Winter Forage

Crop: Winter Forage Acres planted: 42 Plant date: 10/24/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	672.00 *ton*	Dry-weight		52.2	10,800.00	2,400.00	16,400.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	166.40	36.80	249.60	0.00
Total actual harvest content	16.00	165.20	36.71	250.85	1,651.97

Field #3

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 42 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	1,176.00 *ton*	Dry-weight		72.0	12,200.00	3,730.00	22,900.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	56.00	308.00	0.00
Total actual harvest content	28.00	191.30	58.49	359.07	909.44

Field #4

Field name: Field #4

10/25/2022: Winter Forage

Crop: Winter Forage Acres planted: 14 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	224.00 *ton*	Dry-weight		53.6	10,900.00	2,500.00	16,600.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	161.60	36.80	246.40	0.00
Total actual harvest content	16.00	161.84	37.12	246.48	1,618.43

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	392.00 *ton*	Dry-weight		72.2	11,900.00	3,010.00	18,200.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	47.60	280.00	0.00
Total actual harvest content	28.00	185.26	46.86	283.34	1,136.46

Field #5

Field name: Field #5

10/25/2022: Winter Forage

Crop: Winter Forage Acres planted: 25 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	400.00 *ton*	Dry-weight		51.6	10,200.00	2,470.00	17,600.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	156.80	38.40	256.00	0.00
Total actual harvest content	16.00	157.98	38.26	272.59	1,703.68

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	700.00 *ton*	Dry-weight		72.2	11,900.00	3,010.00	18,200.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	47.60	280.00	0.00
Total actual harvest content	28.00	185.26	46.86	283.34	1,136.46

Field #6

Field name: Field #6

10/25/2022: Winter Forage

Crop: Winter Forage Acres planted: 10 Plant date: 10/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	160.00 *ton*	Dry-weight		51.6	10,200.00	2,470.00	17,600.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	156.80	38.40	256.00	0.00
Total actual harvest content	16.00	157.98	38.26	272.59	1,703.68

Field #6

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	280.00 *ton*	Dry-weight		73.0	12,800.00	3,670.00	23,100.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	196.00	53.20	308.00	0.00
Total actual harvest content	28.00	193.54	55.49	349.27	1,013.04

Field #7

Field name: Field #7

10/26/2022: Winter Forage

Crop: Winter Forage Acres planted: 40 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	640.00 *ton*	Dry-weight		51.4	10,700.00	2,440.00	16,900.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	166.40	38.40	262.40	0.00
Total actual harvest content	16.00	166.41	37.95	262.83	1,617.41

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	1,120.00 *ton*	Dry-weight		72.0	14,200.00	3,110.00	19,400.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	47.60	280.00	0.00
Total actual harvest content	28.00	222.66	48.76	304.19	1,176.00

Field #8

Field name: Field #8

10/26/2022: Winter Forage

Crop: Winter Forage Acres planted: 25 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	400.00 *ton*	Dry-weight		51.8	11,100.00	2,420.00	17,400.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	171.20	36.80	265.60	0.00
Total actual harvest content	16.00	171.21	37.33	268.38	1,465.28

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	700.00 *ton*	Dry-weight		73.0	12,800.00	3,670.00	23,100.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	196.00	53.20	280.00	0.00
Total actual harvest content	28.00	193.54	55.49	349.27	1,013.04

Field #9

Field name: Field #9

10/26/2022: Winter Forage

Crop: Winter Forage Acres planted: 38 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	608.00 *ton*	Dry-weight		51.8	11,600.00	2,490.00	18,300.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	179.20	38.40	281.60	0.00
Total actual harvest content	16.00	178.92	38.41	282.26	1,588.67

Field #9

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	1,064.00 *ton*	Dry-weight		63.2	13,500.00	2,810.00	18,000.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	277.20	58.80	313.60	0.00
Total actual harvest content	28.00	278.21	57.91	370.94	1,133.44

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #1 - 10/24/2022: Winter Forage

Field name: Field #1

Crop: Winter Forage — Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	195.02	101.10	288.95	1,914.39	70.00 *ton*
Application event totals		195.02	101.10	288.95	1,914.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	65.45	8.58	69.20	366.94	90,000.00 *gal*
Application event totals		65.45	8.58	69.20	366.94	

Field #1 - 06/01/2023: Corn, silage

Field name: Field #1

Crop: Corn, silage — Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	141.23	38.13	219.45	0.00	98.00 *ton*
Application event totals		141.23	38.13	219.45	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.45	0.00	0.00	0.00	2,220,000.00 *gal*
Application event totals		0.45	0.00	0.00	0.00	

Field #1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.38	0.00	0.00	0.00	1,860,000.00 *gal*
Application event totals		0.38	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	56.33	12.23	45.47	257.50	67,500.00 *gal*
Ag Well Field #3	Ground water	0.29	0.00	0.00	0.00	1,440,000.00 *gal*
Application event totals		56.62	12.23	45.47	257.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	1,500,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	1,500,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	52.85	1.14	45.06	209.76	45,000.00 *gal*
Ag Well Field #3	Ground water	0.29	0.00	0.00	0.00	1,440,000.00 *gal*
Application event totals		53.14	1.14	45.06	209.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	1,500,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Field #10 - 10/27/2022: Winter Forage

Field name: Field #10

Crop: Winter Forage

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	173.35	89.86	256.85	1,701.68	80.00 *ton*
Application event totals		173.35	89.86	256.85	1,701.68	

02/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	50.90	6.68	53.83	285.40	90,000.00 *gal*
Application event totals		50.90	6.68	53.83	285.40	

Field #10 - 06/01/2023: Corn, silage

Field name: Field #10

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	141.23	38.13	219.45	0.00	126.00 *ton*
Application event totals		141.23	38.13	219.45	0.00	

06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.43	0.00	0.00	0.00	2,700,000.00 *gal*
Application event totals		0.43	0.00	0.00	0.00	

Field #10 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.37	0.00	0.00	0.00	2,340,000.00 *gal*
Application event totals		0.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	58.42	12.68	47.15	267.04	90,000.00 *gal*
Ag Well Field #3	Ground water	0.29	0.00	0.00	0.00	1,830,000.00 *gal*
Application event totals		58.70	12.68	47.15	267.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	1,920,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	1,920,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	41.10	0.88	35.05	163.14	45,000.00 *gal*
Ag Well Field #3	Ground water	0.29	0.00	0.00	0.00	1,830,000.00 *gal*
Application event totals		41.39	0.88	35.05	163.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	1,920,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Field #2 - 10/24/2022: Winter Forage

Field name: Field #2

Crop: Winter Forage

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	195.02	101.10	288.95	1,914.39	125.00 *ton*
Application event totals		195.02	101.10	288.95	1,914.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	73.30	9.61	77.51	410.97	180,000.00 *gal*
Application event totals		73.30	9.61	77.51	410.97	

Field #2 - 06/01/2023: Corn, silage

Field name: Field #2

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	141.23	38.13	219.45	0.00	175.00 *ton*
Application event totals		141.23	38.13	219.45	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.38	0.00	0.00	0.00	3,360,000.00 *gal*
Application event totals		0.38	0.00	0.00	0.00	

Field #2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.34	0.00	0.00	0.00	3,000,000.00 *gal*
Application event totals		0.34	0.00	0.00	0.00	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	63.09	13.70	50.92	288.40	135,000.00 *gal*
Ag Well Field #3	Ground water	0.28	0.00	0.00	0.00	2,460,000.00 *gal*
Application event totals		63.37	13.70	50.92	288.40	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	59.19	1.27	50.47	234.93	90,000.00 *gal*
Ag Well Field #3	Ground water	0.28	0.00	0.00	0.00	2,460,000.00 *gal*
Application event totals		59.47	1.27	50.47	234.93	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Field #3 - 10/24/2022: Winter Forage

Field name: Field #3

Crop: Winter Forage

Plant date: 10/24/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	167.16	86.65	247.67	1,640.91	180.00 *ton*
Application event totals		167.16	86.65	247.67	1,640.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.54	7.15	57.67	305.78	225,000.00 *gal*
Application event totals		54.54	7.15	57.67	305.78	

Field #3 - 06/01/2023: Corn, silage

Field name: Field #3

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	141.23	38.13	219.45	0.00	294.00 *ton*
Application event totals		141.23	38.13	219.45	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.34	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		0.34	0.00	0.00	0.00	

Field #3 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.32	0.00	0.00	0.00	4,680,000.00 *gal*
Application event totals		0.32	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	56.33	12.23	45.47	257.50	202,500.00 *gal*
Ag Well Field #3	Ground water	0.28	0.00	0.00	0.00	4,080,000.00 *gal*
Application event totals		56.60	12.23	45.47	257.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.29	0.00	0.00	0.00	4,290,000.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.29	0.00	0.00	0.00	4,290,000.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	46.98	1.01	40.06	186.45	120,000.00 *gal*
Ag Well Field #3	Ground water	0.28	0.00	0.00	0.00	4,080,000.00 *gal*
Application event totals		47.25	1.01	40.06	186.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.29	0.00	0.00	0.00	4,290,000.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Field #4 - 10/25/2022: Winter Forage

Field name: Field #4

Crop: Winter Forage

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	167.16	86.65	247.67	1,640.91	60.00 *ton*
Application event totals		167.16	86.65	247.67	1,640.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	49.09	6.44	51.90	275.21	67,500.00 *gal*
Application event totals		49.09	6.44	51.90	275.21	

Field #4 - 06/01/2023: Corn, silage

Field name: Field #4

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	141.23	38.13	219.45	0.00	98.00 *ton*
Application event totals		141.23	38.13	219.45	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.45	0.00	0.00	0.00	2,220,000.00 *gal*
Application event totals		0.45	0.00	0.00	0.00	

Field #4 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.38	0.00	0.00	0.00	1,860,000.00 *gal*
Application event totals		0.38	0.00	0.00	0.00	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	56.33	12.23	45.47	257.50	67,500.00 *gal*
Ag Well Field #3	Ground water	0.29	0.00	0.00	0.00	1,440,000.00 *gal*
Application event totals		56.62	12.23	45.47	257.50	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	1,500,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	1,500,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	52.85	1.14	45.06	209.76	45,000.00 *gal*
Ag Well Field #3	Ground water	0.29	0.00	0.00	0.00	1,440,000.00 *gal*
Application event totals		53.14	1.14	45.06	209.76	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	1,500,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Field #5 - 10/25/2022: Winter Forage

Field name: Field #5

Crop: Winter Forage

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	156.02	80.88	231.16	1,531.51	100.00 *ton*
Application event totals		156.02	80.88	231.16	1,531.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/11/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.98	7.21	58.13	308.23	135,000.00 *gal*
Application event totals		54.98	7.21	58.13	308.23	

Field #5 - 06/01/2023: Corn, silage

Field name: Field #5

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	141.23	38.13	219.45	0.00	175.00 *ton*
Application event totals		141.23	38.13	219.45	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.38	0.00	0.00	0.00	3,360,000.00 *gal*
Application event totals		0.38	0.00	0.00	0.00	

Field #5 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.34	0.00	0.00	0.00	3,000,000.00 *gal*
Application event totals		0.34	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	63.09	13.70	50.92	288.40	135,000.00 *gal*
Ag Well Field #3	Ground water	0.28	0.00	0.00	0.00	2,460,000.00 *gal*
Application event totals		63.37	13.70	50.92	288.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	88.79	1.91	75.71	352.39	135,000.00 *gal*
Ag Well Field #3	Ground water	0.28	0.00	0.00	0.00	2,460,000.00 *gal*
Application event totals		89.07	1.91	75.71	352.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Field #6 - 10/25/2022: Winter Forage

Field name: Field #6

Crop: Winter Forage

Plant date: 10/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	156.02	80.88	231.16	1,531.51	40.00 *ton*
Application event totals		156.02	80.88	231.16	1,531.51	

02/11/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	45.81	6.01	48.44	256.86	45,000.00 *gal*
Application event totals		45.81	6.01	48.44	256.86	

Field #6 - 06/01/2023: Corn, silage

Field name: Field #6

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	141.23	38.13	219.45	0.00	70.00 *ton*
Application event totals		141.23	38.13	219.45	0.00	

06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.51	0.00	0.00	0.00	1,800,000.00 *gal*
Application event totals		0.51	0.00	0.00	0.00	

Field #6 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.43	0.00	0.00	0.00	1,500,000.00 *gal*
Application event totals		0.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	52.57	11.42	42.43	240.34	45,000.00 *gal*
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	1,050,000.00 *gal*
Application event totals		52.87	11.42	42.43	240.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	37.33	0.00	0.00	0.00	1,110,000.00 *gal*
Application event totals		37.33	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.31	0.00	0.00	0.00	1,110,000.00 *gal*
Application event totals		0.31	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	35.31	0.00	0.00	0.00	1,050,000.00 *gal*
Application event totals		35.31	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.31	0.00	0.00	0.00	1,110,000.00 *gal*
Application event totals		0.31	0.00	0.00	0.00	

Field #7 - 10/26/2022: Winter Forage

Field name: Field #7

Crop: Winter Forage

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	156.02	80.88	231.16	1,531.51	160.00 *ton*
Application event totals		156.02	80.88	231.16	1,531.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/11/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	51.54	6.76	54.50	288.97	202,500.00 *gal*
Application event totals		51.54	6.76	54.50	288.97	

Field #7 - 06/01/2023: Corn, silage

Field name: Field #7

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	141.23	38.13	219.45	0.00	280.00 *ton*
Application event totals		141.23	38.13	219.45	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.34	0.00	0.00	0.00	4,800,000.00 *gal*
Application event totals		0.34	0.00	0.00	0.00	

Field #7 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	35.73	0.00	0.00	0.00	4,250,000.00 *gal*
Application event totals		35.73	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	59.15	12.84	47.74	270.38	202,500.00 *gal*
Ag Well Field #3	Ground water	0.28	0.00	0.00	0.00	3,930,000.00 *gal*
Application event totals		59.42	12.84	47.74	270.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	34.81	0.00	0.00	0.00	4,140,000.00 *gal*
Application event totals		34.81	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.29	0.00	0.00	0.00	4,140,000.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	33.04	0.00	0.00	0.00	3,930,000.00 *gal*
Application event totals		33.04	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.29	0.00	0.00	0.00	4,140,000.00 *gal*
Application event totals		0.29	0.00	0.00	0.00	

Field #8 - 10/26/2022: Winter Forage

Field name: Field #8

Crop: Winter Forage

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	156.02	80.88	231.16	1,531.51	100.00 *ton*
Application event totals		156.02	80.88	231.16	1,531.51	

02/11/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	54.98	7.21	58.13	308.23	135,000.00 *gal*
Application event totals		54.98	7.21	58.13	308.23	

Field #8 - 06/01/2023: Corn, silage

Field name: Field #8

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	141.23	38.13	219.45	0.00	175.00 *ton*
Application event totals		141.23	38.13	219.45	0.00	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.38	0.00	0.00	0.00	3,360,000.00 *gal*
Application event totals		0.38	0.00	0.00	0.00	

Field #8 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	36.99	0.00	0.00	0.00	2,750,000.00 *gal*
Application event totals		36.99	0.00	0.00	0.00	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	42.06	9.13	33.95	192.27	90,000.00 *gal*
Ag Well Field #3	Ground water	0.28	0.00	0.00	0.00	2,460,000.00 *gal*
Application event totals		42.34	9.13	33.95	192.27	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	35.11	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		35.11	0.00	0.00	0.00	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	33.09	0.00	0.00	0.00	2,460,000.00 *gal*
Application event totals		33.09	0.00	0.00	0.00	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #3	Ground water	0.30	0.00	0.00	0.00	2,610,000.00 *gal*
Application event totals		0.30	0.00	0.00	0.00	

Field #9 - 10/26/2022: Winter Forage

Field name: Field #9

Crop: Winter Forage

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	164.23	85.13	243.33	1,612.12	160.00 *ton*
Application event totals		164.23	85.13	243.33	1,612.12	

02/11/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	48.23	6.32	50.99	270.38	180,000.00 *gal*
Application event totals		48.23	6.32	50.99	270.38	

Field #9 - 06/01/2023: Corn, silage

Field name: Field #9

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	141.23	38.13	219.45	0.00	266.00 *ton*
Application event totals		141.23	38.13	219.45	0.00	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	40.89	0.00	0.00	0.00	4,620,000.00 *gal*
Application event totals		40.89	0.00	0.00	0.00	

Field #9 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	38.23	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		38.23	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	55.34	12.02	44.67	252.99	180,000.00 *gal*
Ag Well Field #6	Ground water	32.92	0.00	0.00	0.00	3,720,000.00 *gal*
Application event totals		88.26	12.02	44.67	252.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	34.52	0.00	0.00	0.00	3,900,000.00 *gal*
Application event totals		34.52	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	34.52	0.00	0.00	0.00	3,900,000.00 *gal*
Application event totals		34.52	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well Field #6	Ground water	32.92	0.00	0.00	0.00	3,720,000.00 *gal*
Application event totals		32.92	0.00	0.00	0.00	

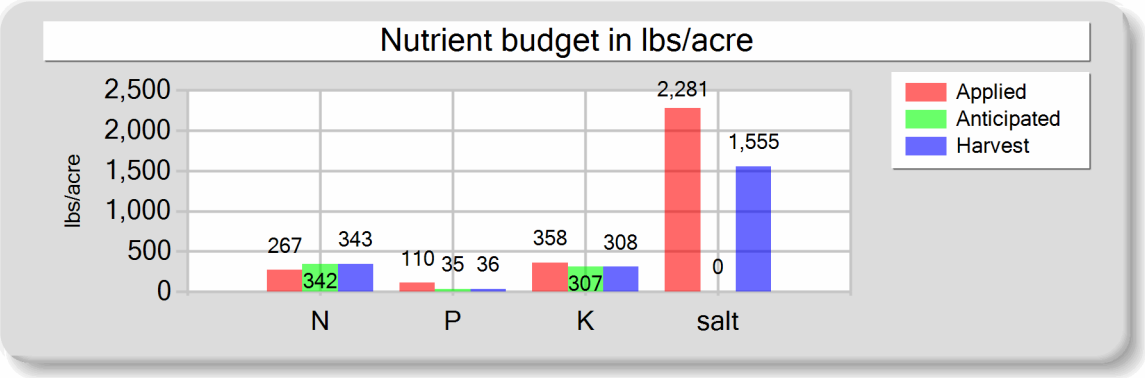
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	77.88	1.68	66.41	309.12	180,000.00 *gal*
Ag Well Field #6	Ground water	34.52	0.00	0.00	0.00	3,900,000.00 *gal*
Application event totals		112.40	1.68	66.41	309.12	

B. NUTRIENT BUDGET

Field #1 - 10/24/2022: Winter Forage

Field name: Field #1 Crop: Winter Forage Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	195.02	101.10	288.95	1,914.39
Process wastewater	65.45	8.58	69.20	366.94
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.47	109.68	358.15	2,281.33
Anticipated crop nutrient removal	342.40	35.20	307.20	0.00
Actual crop nutrient removal	342.84	35.87	307.92	1,555.46
Nutrient balance	-75.36	73.81	50.24	725.88
Applied to removed ratio	0.78	3.06	1.16	1.47

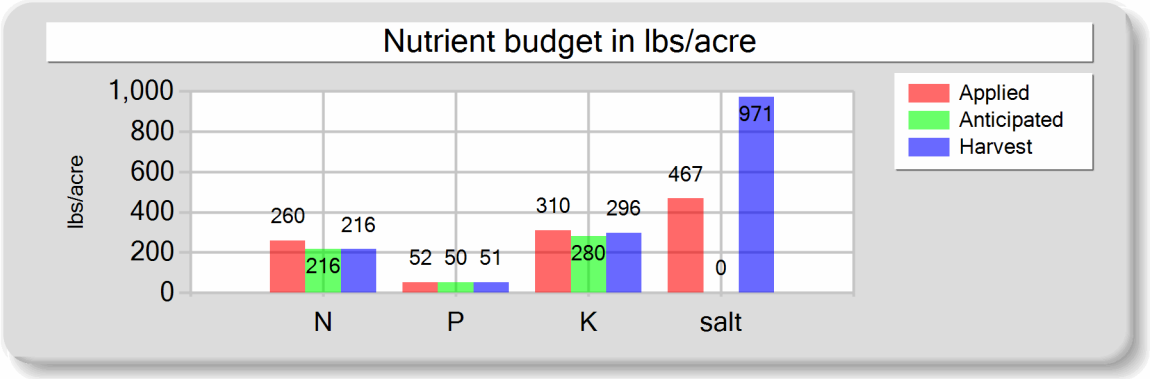
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
90,000.00 *gallons*
3.31 *acre-inches*
0.24 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #1 - 06/01/2023: Corn, silage

Field name: Field #1 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.23	38.13	219.45	0.00
Process wastewater	109.18	13.37	90.53	467.26
Fresh water	2.32	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.73	51.50	309.98	467.26
Anticipated crop nutrient removal	215.60	50.40	280.00	0.00
Actual crop nutrient removal	215.68	51.20	296.35	971.38
Nutrient balance	44.05	0.30	13.63	-504.12
Applied to removed ratio	1.20	1.01	1.05	0.48

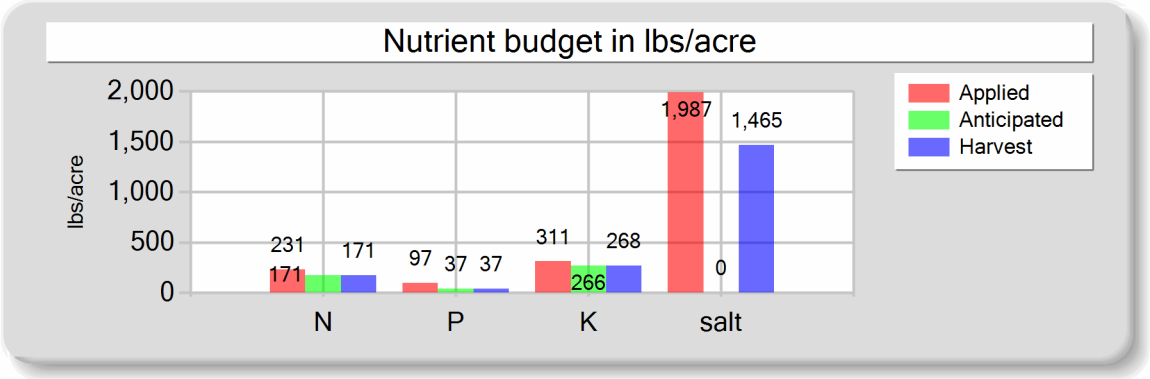
Fresh water applied	
11,460,000.00	*gallons*
422.03	*acre-inches*
30.15	*inches/acre*

Process wastewater applied	
112,500.00	*gallons*
4.14	*acre-inches*
0.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #10 - 10/27/2022: Winter Forage

Field name: Field #10 Crop: Winter Forage Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	173.35	89.86	256.85	1,701.68
Process wastewater	50.90	6.68	53.83	285.40
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.26	96.54	310.67	1,987.08
Anticipated crop nutrient removal	171.20	36.80	265.60	0.00
Actual crop nutrient removal	171.21	37.33	268.38	1,465.28
Nutrient balance	60.05	59.21	42.29	521.80
Applied to removed ratio	1.35	2.59	1.16	1.36

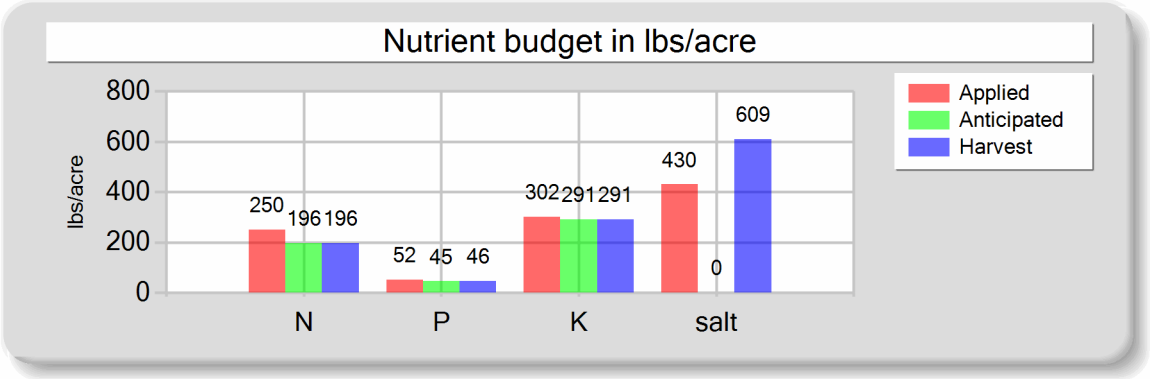
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
90,000.00 *gallons*
3.31 *acre-inches*
0.18 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #10 - 06/01/2023: Corn, silage

Field name: Field #10 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.23	38.13	219.45	0.00
Process wastewater	99.52	13.57	82.20	430.18
Fresh water	2.28	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.03	51.70	301.65	430.18
Anticipated crop nutrient removal	196.00	44.80	291.20	0.00
Actual crop nutrient removal	195.92	46.10	291.41	609.17
Nutrient balance	54.11	5.60	10.24	-178.98
Applied to removed ratio	1.28	1.12	1.04	0.71

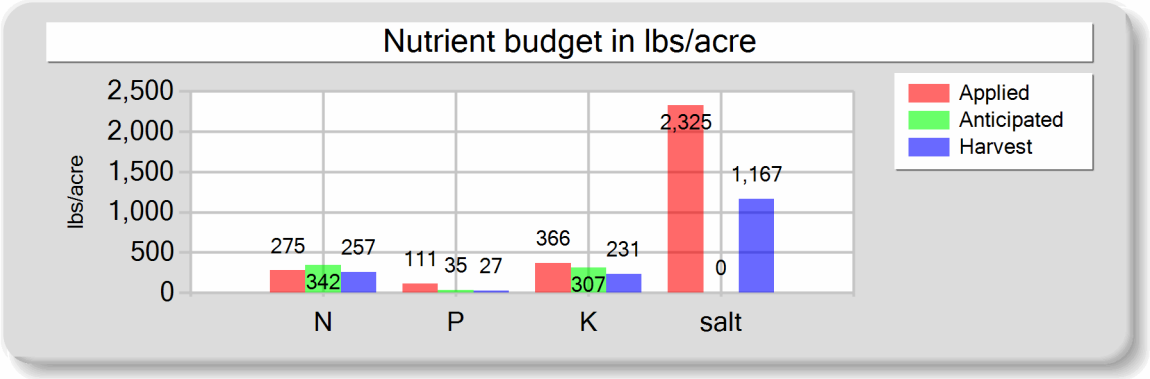
Fresh water applied	
14,460,000.00	*gallons*
532.51	*acre-inches*
29.58	*inches/acre*

Process wastewater applied	
135,000.00	*gallons*
4.97	*acre-inches*
0.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #2 - 10/24/2022: Winter Forage

Field name: Field #2 Crop: Winter Forage Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	195.02	101.10	288.95	1,914.39
Process wastewater	73.30	9.61	77.51	410.97
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.33	110.71	366.46	2,325.36
Anticipated crop nutrient removal	342.40	35.20	307.20	0.00
Actual crop nutrient removal	257.13	26.90	230.94	1,166.59
Nutrient balance	18.20	83.81	135.52	1,158.77
Applied to removed ratio	1.07	4.12	1.59	1.99

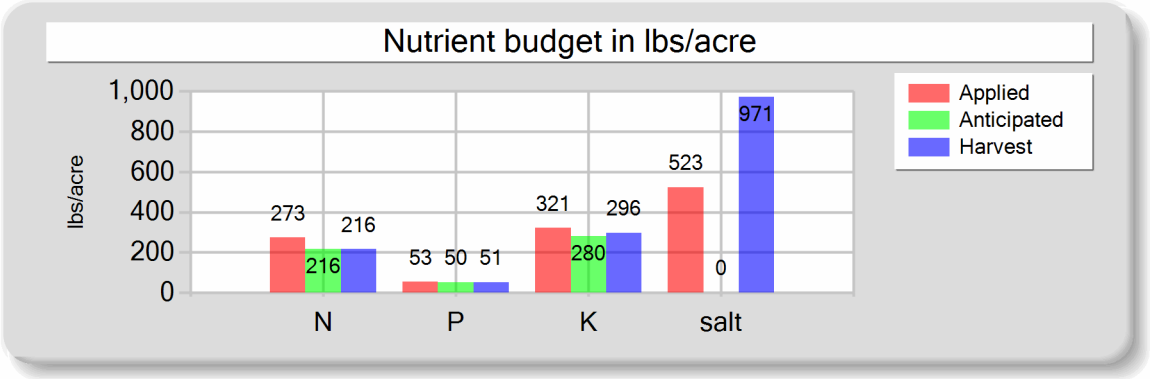
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
180,000.00	*gallons*
6.63	*acre-inches*
0.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #2 - 06/01/2023: Corn, silage

Field name: Field #2 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.23	38.13	219.45	0.00
Process wastewater	122.28	14.97	101.39	523.33
Fresh water	2.17	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	272.68	53.11	320.84	523.33
Anticipated crop nutrient removal	215.60	50.40	280.00	0.00
Actual crop nutrient removal	215.68	51.20	296.35	971.38
Nutrient balance	57.00	1.90	24.49	-448.04
Applied to removed ratio	1.26	1.04	1.08	0.54

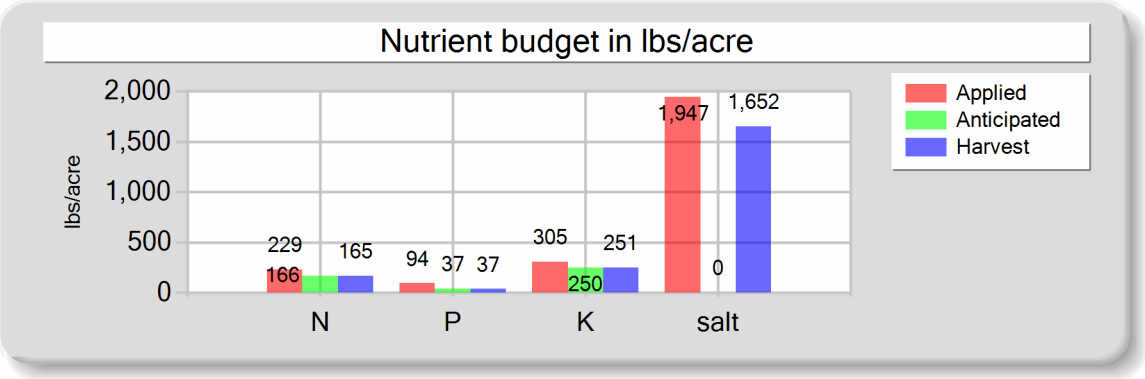
Fresh water applied	
19,110,000.00	*gallons*
703.76	*acre-inches*
28.15	*inches/acre*

Process wastewater applied	
225,000.00	*gallons*
8.29	*acre-inches*
0.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #3 - 10/24/2022: Winter Forage

Field name: Field #3 Crop: Winter Forage Plant date: 10/24/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	167.16	86.65	247.67	1,640.91
Process wastewater	54.54	7.15	57.67	305.78
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.70	93.81	305.34	1,946.69
Anticipated crop nutrient removal	166.40	36.80	249.60	0.00
Actual crop nutrient removal	165.20	36.71	250.85	1,651.97
Nutrient balance	63.51	57.10	54.49	294.72
Applied to removed ratio	1.38	2.56	1.22	1.18

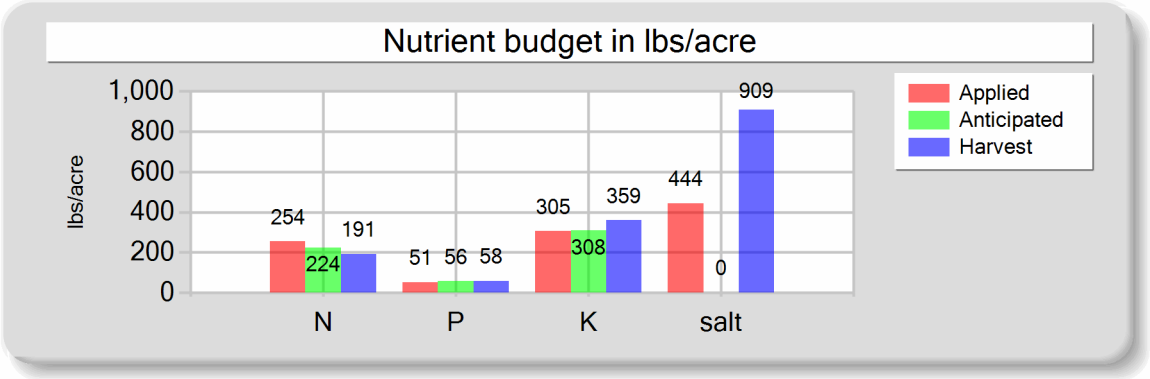
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
225,000.00 *gallons*
8.29 *acre-inches*
0.20 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #3 - 06/01/2023: Corn, silage

Field name: Field #3 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.23	38.13	219.45	0.00
Process wastewater	103.31	13.24	85.52	443.95
Fresh water	2.08	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	253.61	51.37	304.97	443.95
Anticipated crop nutrient removal	224.00	56.00	308.00	0.00
Actual crop nutrient removal	191.30	58.49	359.07	909.44
Nutrient balance	62.32	-7.11	-54.10	-465.49
Applied to removed ratio	1.33	0.88	0.85	0.49

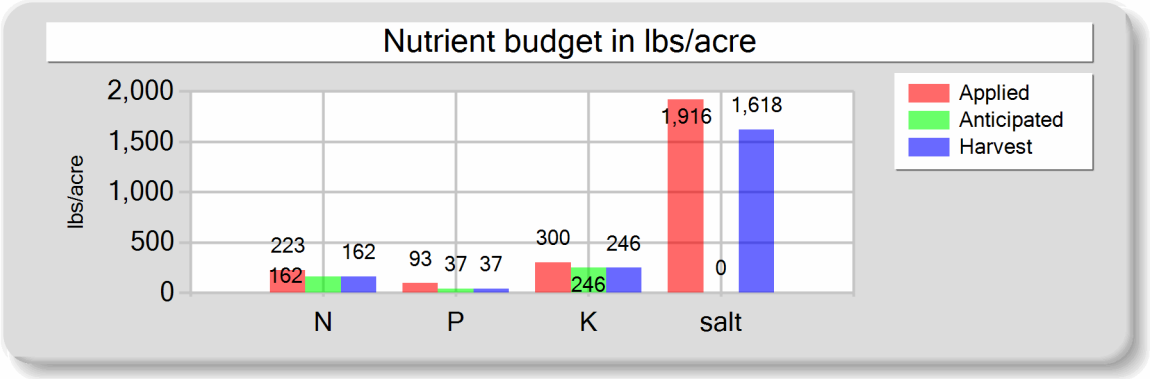
Fresh water applied	
30,750,000.00	*gallons*
1,132.42	*acre-inches*
26.96	*inches/acre*

Process wastewater applied	
322,500.00	*gallons*
11.88	*acre-inches*
0.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #4 - 10/25/2022: Winter Forage

Field name: Field #4 Crop: Winter Forage Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	167.16	86.65	247.67	1,640.91
Process wastewater	49.09	6.44	51.90	275.21
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.25	93.09	299.58	1,916.11
Anticipated crop nutrient removal	161.60	36.80	246.40	0.00
Actual crop nutrient removal	161.84	37.12	246.48	1,618.43
Nutrient balance	61.41	55.97	53.10	297.68
Applied to removed ratio	1.38	2.51	1.22	1.18

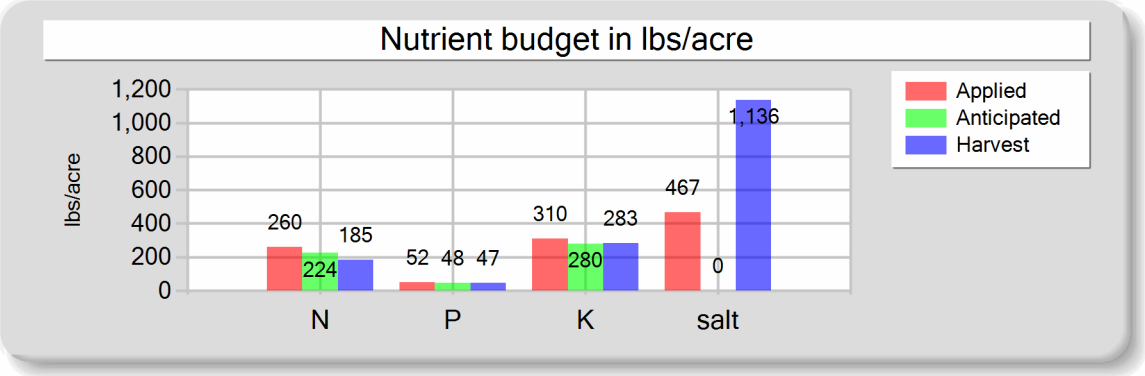
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
67,500.00 *gallons*
2.49 *acre-inches*
0.18 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #4 - 06/01/2023: Corn, silage

Field name: Field #4 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.23	38.13	219.45	0.00
Process wastewater	109.18	13.37	90.53	467.26
Fresh water	2.32	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.73	51.50	309.98	467.26
Anticipated crop nutrient removal	224.00	47.60	280.00	0.00
Actual crop nutrient removal	185.26	46.86	283.34	1,136.46
Nutrient balance	74.47	4.64	26.64	-669.20
Applied to removed ratio	1.40	1.10	1.09	0.41

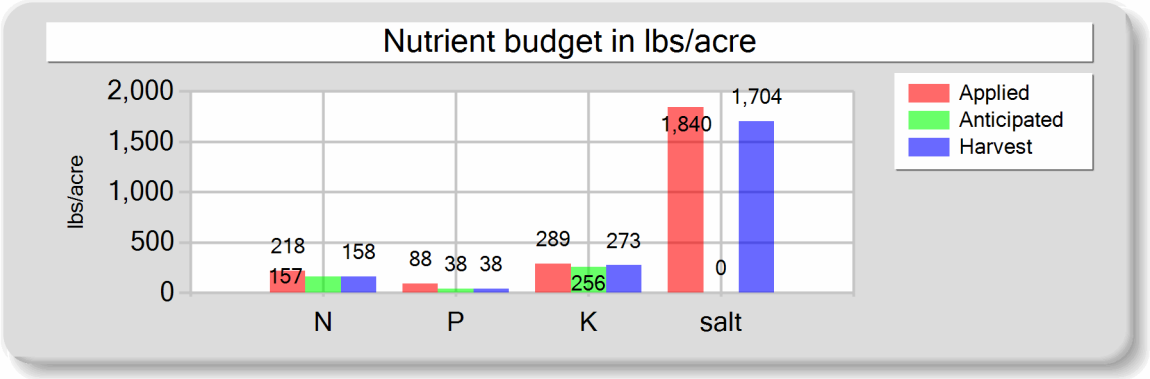
Fresh water applied	
11,460,000.00	*gallons*
422.03	*acre-inches*
30.15	*inches/acre*

Process wastewater applied	
112,500.00	*gallons*
4.14	*acre-inches*
0.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #5 - 10/25/2022: Winter Forage

Field name: Field #5 Crop: Winter Forage Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	156.02	80.88	231.16	1,531.51
Process wastewater	54.98	7.21	58.13	308.23
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.00	88.09	289.29	1,839.74
Anticipated crop nutrient removal	156.80	38.40	256.00	0.00
Actual crop nutrient removal	157.98	38.26	272.59	1,703.68
Nutrient balance	60.02	49.83	16.70	136.06
Applied to removed ratio	1.38	2.30	1.06	1.08

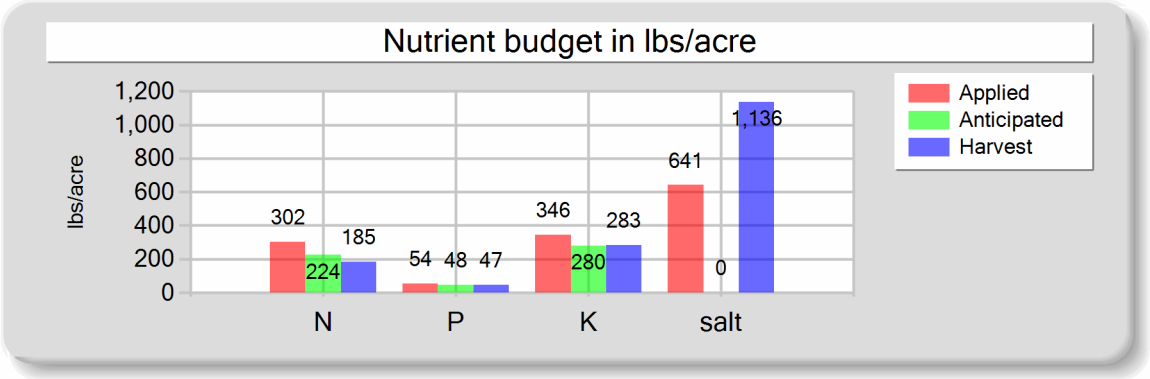
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
135,000.00	*gallons*
4.97	*acre-inches*
0.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #5 - 06/01/2023: Corn, silage

Field name: Field #5 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.23	38.13	219.45	0.00
Process wastewater	151.87	15.61	126.63	640.80
Fresh water	2.17	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.27	53.74	346.08	640.80
Anticipated crop nutrient removal	224.00	47.60	280.00	0.00
Actual crop nutrient removal	185.26	46.86	283.34	1,136.46
Nutrient balance	117.02	6.88	62.74	-495.67
Applied to removed ratio	1.63	1.15	1.22	0.56

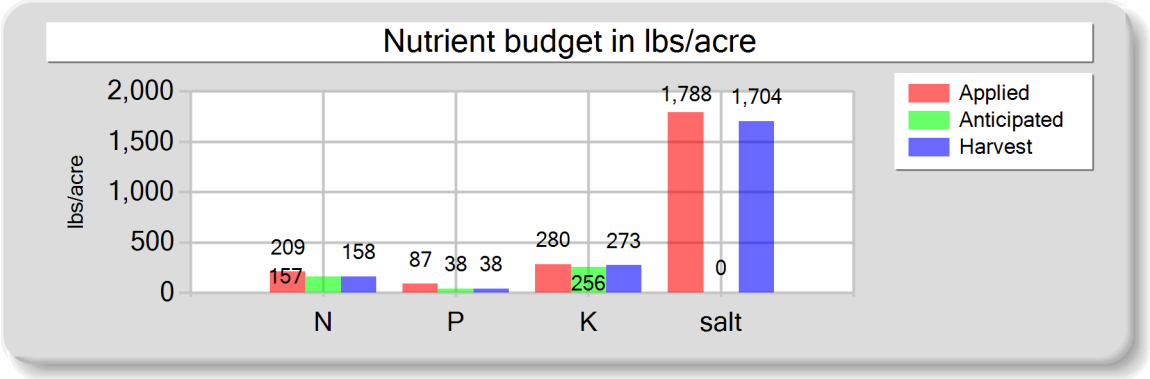
Fresh water applied	
19,110,000.00	*gallons*
703.76	*acre-inches*
28.15	*inches/acre*

Process wastewater applied	
270,000.00	*gallons*
9.94	*acre-inches*
0.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #6 - 10/25/2022: Winter Forage

Field name: Field #6 Crop: Winter Forage Plant date: 10/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	156.02	80.88	231.16	1,531.51
Process wastewater	45.81	6.01	48.44	256.86
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.83	86.89	279.60	1,788.37
Anticipated crop nutrient removal	156.80	38.40	256.00	0.00
Actual crop nutrient removal	157.98	38.26	272.59	1,703.68
Nutrient balance	50.86	48.63	7.01	84.69
Applied to removed ratio	1.32	2.27	1.03	1.05

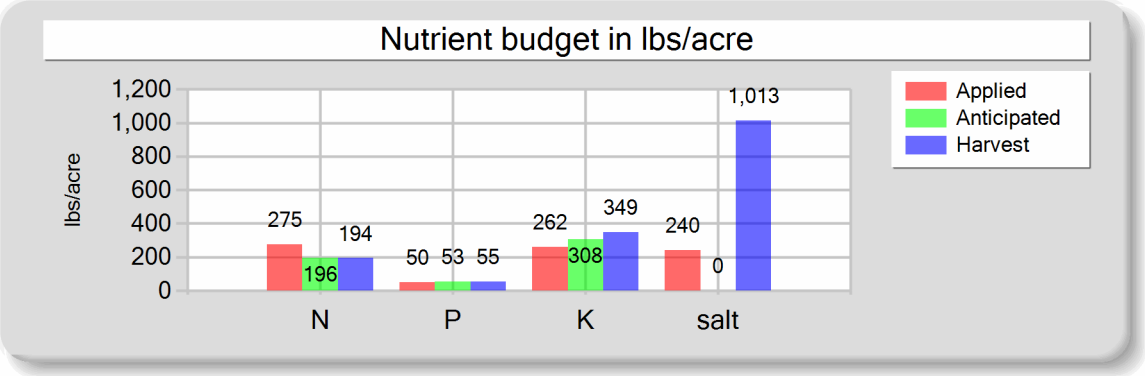
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
45,000.00 *gallons*
1.66 *acre-inches*
0.17 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #6 - 06/01/2023: Corn, silage

Field name: Field #6 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.23	38.13	219.45	0.00
Process wastewater	52.57	11.42	42.43	240.34
Fresh water	74.51	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.31	49.55	261.89	240.34
Anticipated crop nutrient removal	196.00	53.20	308.00	0.00
Actual crop nutrient removal	193.54	55.49	349.27	1,013.04
Nutrient balance	81.78	-5.94	-87.38	-772.70
Applied to removed ratio	1.42	0.89	0.75	0.24

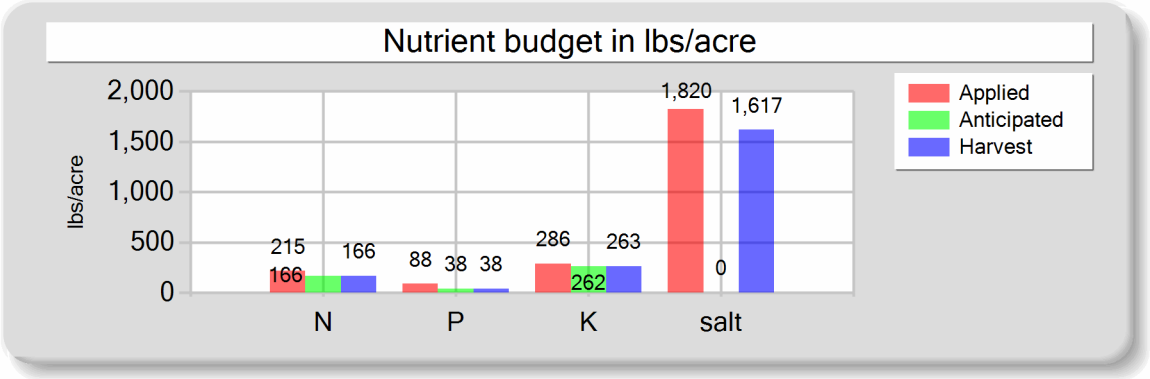
Fresh water applied	
8,730,000.00	*gallons*
321.50	*acre-inches*
32.15	*inches/acre*

Process wastewater applied	
45,000.00	*gallons*
1.66	*acre-inches*
0.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #7 - 10/26/2022: Winter Forage

Field name: Field #7 Crop: Winter Forage Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	156.02	80.88	231.16	1,531.51
Process wastewater	51.54	6.76	54.50	288.97
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.56	87.64	285.66	1,820.48
Anticipated crop nutrient removal	166.40	38.40	262.40	0.00
Actual crop nutrient removal	166.41	37.95	262.83	1,617.41
Nutrient balance	48.15	49.69	22.83	203.07
Applied to removed ratio	1.29	2.31	1.09	1.13

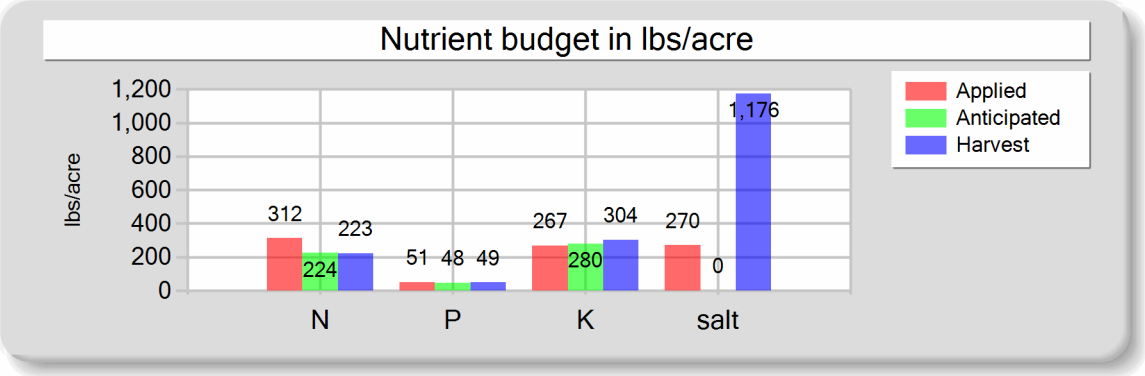
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
202,500.00 *gallons*
7.46 *acre-inches*
0.19 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #7 - 06/01/2023: Corn, silage

Field name: Field #7 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.23	38.13	219.45	0.00
Process wastewater	59.15	12.84	47.74	270.38
Fresh water	104.79	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	312.17	50.98	267.19	270.38
Anticipated crop nutrient removal	224.00	47.60	280.00	0.00
Actual crop nutrient removal	222.66	48.76	304.19	1,176.00
Nutrient balance	89.51	2.21	-37.00	-905.62
Applied to removed ratio	1.40	1.05	0.88	0.23

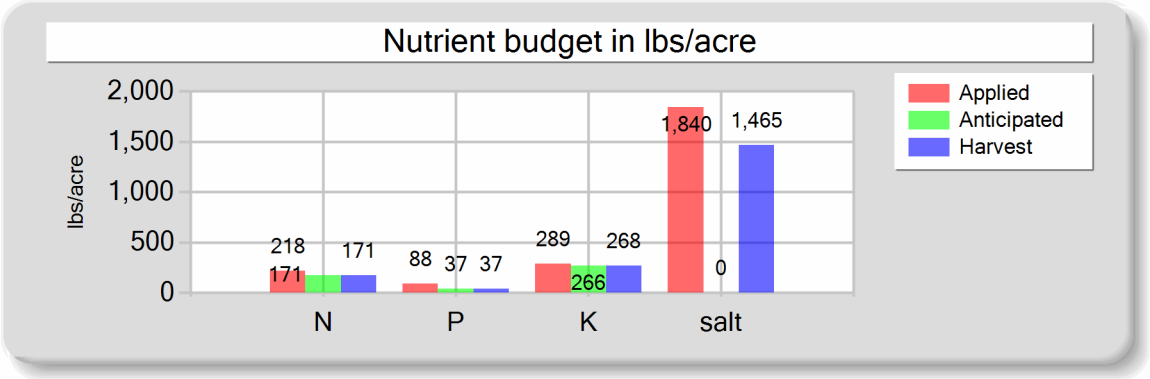
Fresh water applied	
29,330,000.00	*gallons*
1,080.12	*acre-inches*
27.00	*inches/acre*

Process wastewater applied	
202,500.00	*gallons*
7.46	*acre-inches*
0.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #8 - 10/26/2022: Winter Forage

Field name: Field #8 Crop: Winter Forage Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	156.02	80.88	231.16	1,531.51
Process wastewater	54.98	7.21	58.13	308.23
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.00	88.09	289.29	1,839.74
Anticipated crop nutrient removal	171.20	36.80	265.60	0.00
Actual crop nutrient removal	171.21	37.33	268.38	1,465.28
Nutrient balance	46.79	50.76	20.91	374.46
Applied to removed ratio	1.27	2.36	1.08	1.26

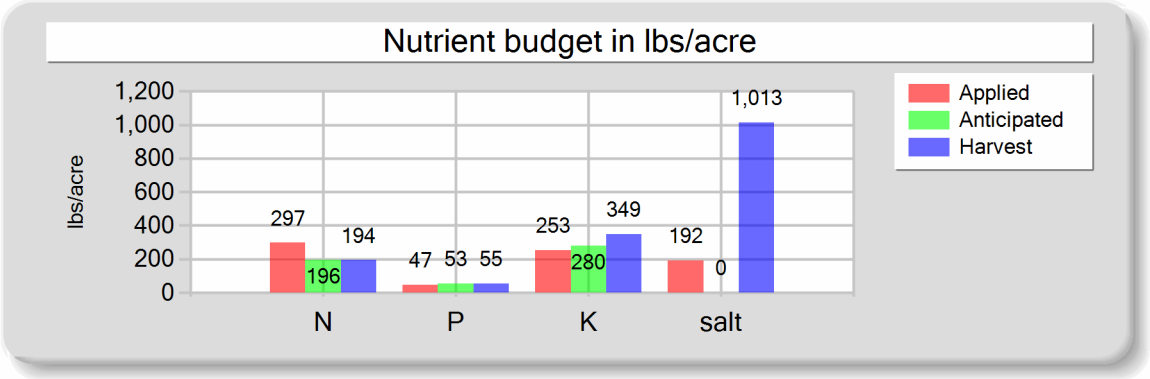
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
135,000.00 *gallons*
4.97 *acre-inches*
0.20 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #8 - 06/01/2023: Corn, silage

Field name: Field #8 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.23	38.13	219.45	0.00
Process wastewater	42.06	9.13	33.95	192.27
Fresh water	106.45	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	296.74	47.27	253.40	192.27
Anticipated crop nutrient removal	196.00	53.20	280.00	0.00
Actual crop nutrient removal	193.54	55.49	349.27	1,013.04
Nutrient balance	103.20	-8.22	-95.87	-820.77
Applied to removed ratio	1.53	0.85	0.73	0.19

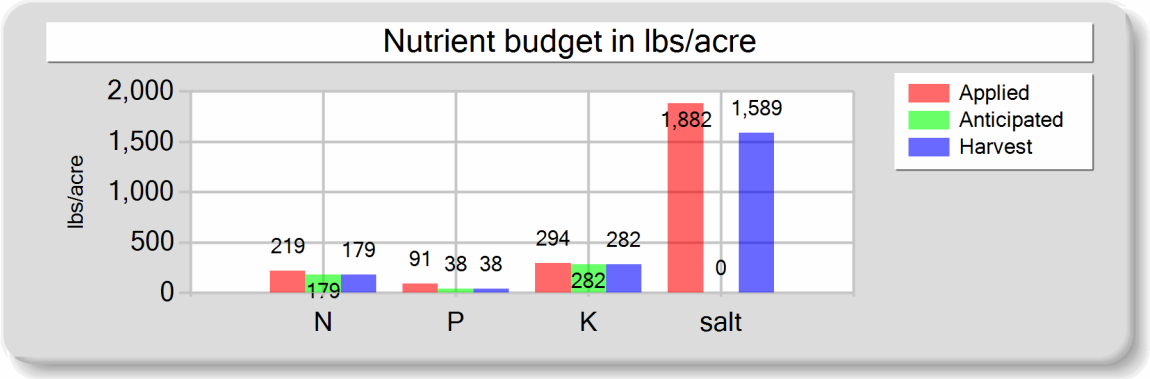
Fresh water applied	
18,860,000.00	*gallons*
694.55	*acre-inches*
27.78	*inches/acre*

Process wastewater applied	
90,000.00	*gallons*
3.31	*acre-inches*
0.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #9 - 10/26/2022: Winter Forage

Field name: Field #9 Crop: Winter Forage Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	164.23	85.13	243.33	1,612.12
Process wastewater	48.23	6.32	50.99	270.38
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.46	91.46	294.32	1,882.50
Anticipated crop nutrient removal	179.20	38.40	281.60	0.00
Actual crop nutrient removal	178.92	38.41	282.26	1,588.67
Nutrient balance	40.54	53.05	12.06	293.82
Applied to removed ratio	1.23	2.38	1.04	1.18

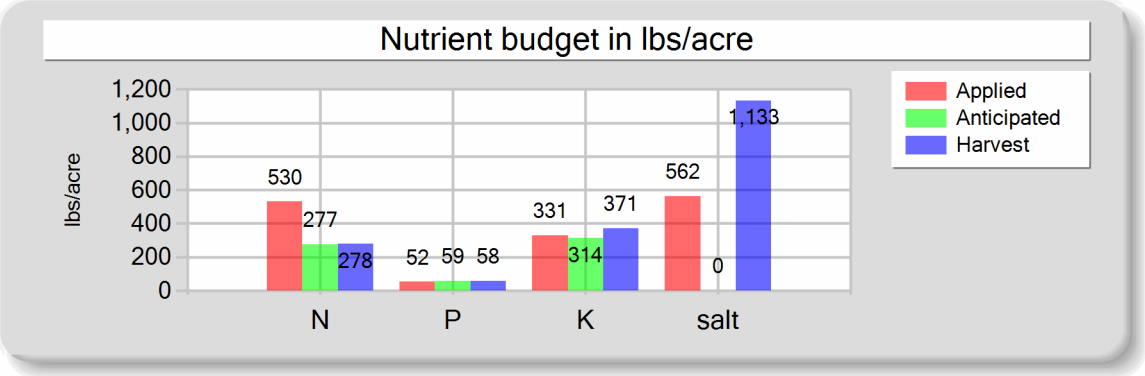
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
180,000.00 *gallons*
6.63 *acre-inches*
0.17 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #9 - 06/01/2023: Corn, silage

Field name: Field #9 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	141.23	38.13	219.45	0.00
Process wastewater	133.22	13.69	111.08	562.10
Fresh water	248.51	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	529.97	51.83	330.53	562.10
Anticipated crop nutrient removal	277.20	58.80	313.60	0.00
Actual crop nutrient removal	278.21	57.91	370.94	1,133.44
Nutrient balance	251.76	-6.08	-40.41	-571.34
Applied to removed ratio	1.90	0.89	0.89	0.50

Fresh water applied	
28,080,000.00	*gallons*
1,034.09	*acre-inches*
27.21	*inches/acre*

Process wastewater applied	
360,000.00	*gallons*
13.26	*acre-inches*
0.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2348030A

Sample and source description: T2348030A

Sample date: 12/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 28.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,200.00	14,100.00	40,300.00	27,800.00	16,300.00	5,850.00	10,800.00	1,930.00		26.70
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3130029B

Sample and source description: T3130029B

Sample date: 05/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 22.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,000.00	3,510.00	20,200.00							0.00
DL	1.00	1.00	1.00							1.00

T3264042B

Sample and source description: T3264042B

Sample date: 09/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	4,870.00	24,800.00	10,600.00	7,780.00	5,280.00	6,070.00	910.00		52.30
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3059013A

Sample and source description: T3059013A

Sample date: 02/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,220.00	767.00	0.00	0.00	160.00	1,290.00								10,800.00	6,840
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3130029A

Sample and source description: T3130029A

Sample date: 05/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,400.00	792.00	0.00	0.00	304.00	1,130.00								11,900.00	6,400
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3264042A

Sample and source description: T3264042A

Sample date: 09/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,970.00	829.00	0.00	0.26	42.40	1,680.00	37.20	58.10	196.00	6,960.00	0.00	357.00	484.00	11,600.00	7,820
DL	1.00	1.00	1.00	0.10	1.00	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

C. FRESH WATER ANALYSES

Ag Well Field #3

Ag Well Field #3

T3314020C

Sample description: T3314020C

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.34	0.00	0.34								265.00	0
DL	0.10	0.50	0.10								100.00	1

Ag Well Field #6

T3314020D

Sample description: T3314020D

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	40.30	0.00	40.30								1,280.00	0
DL	1.00	0.50	1.00								100.00	1

Dairy/Dom. #1

T3314020A

Sample description: T3314020A

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.09	0.00	4.09								474.00	0
DL	1.00	0.50	1.00								100.00	1

Dairy/Dom. #2

Dairy/Dom. #2

T3314020B

Sample description: T3314020B

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.50	0.00	8.50								650.00	0
DL	1.00	0.50	1.00								100.00	1

D. SOIL ANALYSES

Field #1

T3349019A

Sample and source description: T3349019A

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			123.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #10

T3349019G

Sample and source description: T3349019G

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			88.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #2

Field #2

T3349019A

Sample and source description: T3349019A

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			123.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #3

T3349019B

Sample and source description: T3349019B

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #4

T3349019C

Sample and source description: T3349019C

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			108.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #5

Field #5

T3349019C

Sample and source description: T3349019C

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			108.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #6

T3349019D

Sample and source description: T3349019D

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #7

T3349019E

Sample and source description: T3349019E

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #8

Field #8

T3349019D

Sample and source description: T3349019D

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field #9

T3349019F

Sample and source description: T3349019F

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00		0.00	0.00	
DL			1.00		1.00	1.00	

E. PLANT TISSUE ANALYSES

Field #1 - 10/24/2022: Winter Forage

T3139077A

Sample and source description: T3139077A

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,600.00	2,260.00	19,400.00		9.80
DL	1.00	1.00	1.00		1.00

Field #1 - 06/01/2023: Corn, silage

Field #1 - 06/01/2023: Corn, silage

T3272039A

Sample and source description: T3272039A

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	3,110.00	18,000.00		5.90
DL	1.00	1.00	1.00		1.00

Field #10 - 10/27/2022: Winter Forage

T3139077F

Sample and source description: T3139077F

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	2,420.00	17,400.00		9.50
DL	1.00	1.00	1.00		1.00

Field #10 - 06/01/2023: Corn, silage

T3272039G

Sample and source description: T3272039G

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,900.00	2,800.00	17,700.00		3.70
DL	1.00	1.00	1.00		1.00

Field #2 - 10/24/2022: Winter Forage

T3139077A

Sample and source description: T3139077A

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,600.00	2,260.00	19,400.00		9.80
DL	1.00	1.00	1.00		1.00

Field #2 - 06/01/2023: Corn, silage

T3272039A

Sample and source description: T3272039A

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	3,110.00	18,000.00		5.90
DL	1.00	1.00	1.00		1.00

Field #3 - 10/24/2022: Winter Forage

T3139077B

Sample and source description: T3139077B

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	2,400.00	16,400.00		10.80
DL	1.00	1.00	1.00		1.00

Field #3 - 06/01/2023: Corn, silage

T3272039B

Sample and source description: T3272039B

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	3,730.00	22,900.00		5.80
DL	1.00	1.00	1.00		1.00

Field #4 - 10/25/2022: Winter Forage

T3139077C

Sample and source description: T3139077C

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	2,500.00	16,600.00		10.90
DL	1.00	1.00	1.00		1.00

Field #4 - 06/01/2023: Corn, silage

T3272039C

Sample and source description: T3272039C

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,900.00	3,010.00	18,200.00		7.30
DL	1.00	1.00	1.00		1.00

Field #5 - 10/25/2022: Winter Forage

T3139077D

Sample and source description: T3139077D

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	2,470.00	17,600.00		11.00
DL	1.00	1.00	1.00		1.00

Field #5 - 06/01/2023: Corn, silage

T3272039C

Sample and source description: T3272039C

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,900.00	3,010.00	18,200.00		7.30
DL	1.00	1.00	1.00		1.00

Field #6 - 10/25/2022: Winter Forage

T3139077D

Sample and source description: T3139077D

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	2,470.00	17,600.00		11.00
DL	1.00	1.00	1.00		1.00

Field #6 - 06/01/2023: Corn, silage

T3272039E

Sample and source description: T3272039E

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	3,670.00	23,100.00		6.70
DL	1.00	1.00	1.00		1.00

Field #7 - 10/26/2022: Winter Forage

T3139077E

Sample and source description: T3139077E

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	2,440.00	16,900.00		10.40
DL	1.00	1.00	1.00		1.00

Field #7 - 06/01/2023: Corn, silage

T3272039D

Sample and source description: T3272039D

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	3,110.00	19,400.00		7.50
DL	1.00	1.00	1.00		1.00

Field #8 - 10/26/2022: Winter Forage

T3139077F

Sample and source description: T3139077F

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	2,420.00	17,400.00		9.50
DL	1.00	1.00	1.00		1.00

Field #8 - 06/01/2023: Corn, silage

T3272039E

Sample and source description: T3272039E

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	3,670.00	23,100.00		6.70
DL	1.00	1.00	1.00		1.00

Field #9 - 10/26/2022: Winter Forage

T3139077G

Sample and source description: T3139077G

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	2,490.00	18,300.00		10.30
DL	1.00	1.00	1.00		1.00

Field #9 - 06/01/2023: Corn, silage

T3272039F

Sample and source description: T3272039F

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	2,810.00	18,000.00		5.50
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

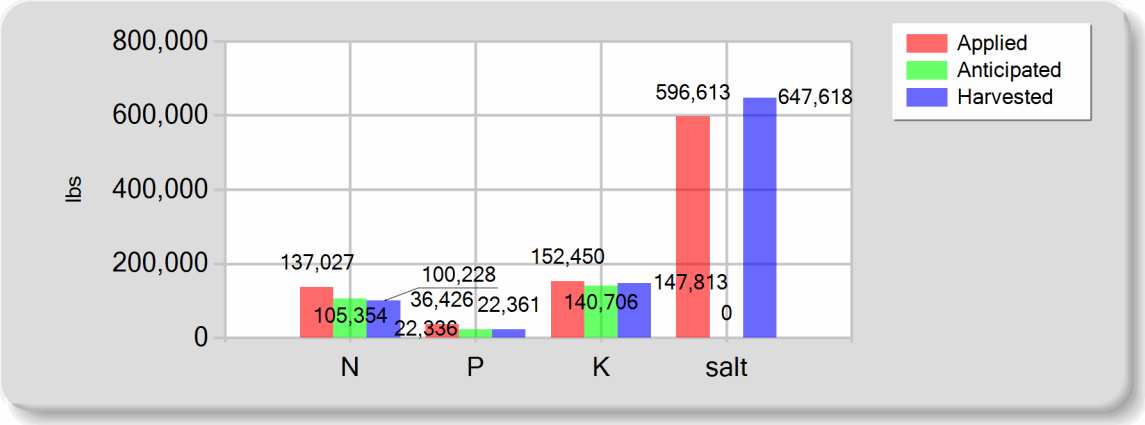
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

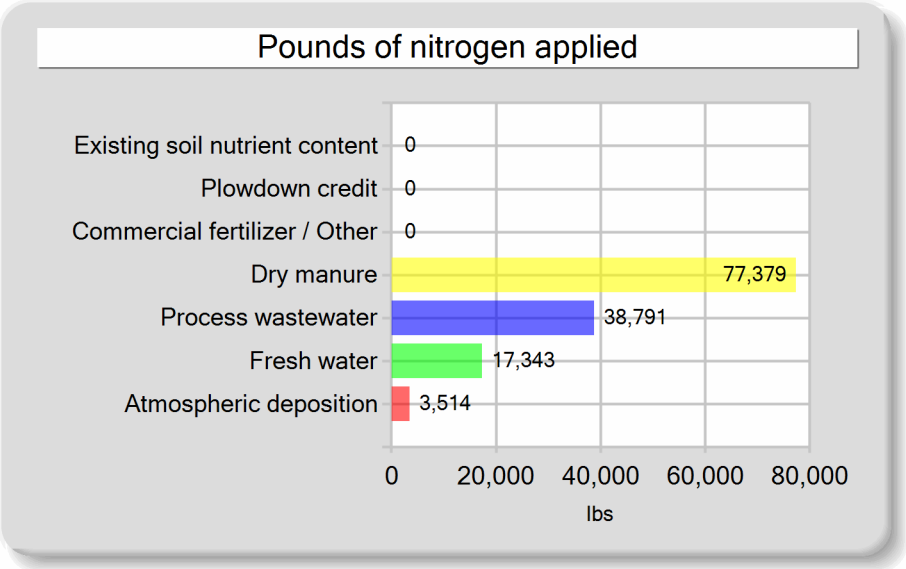
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

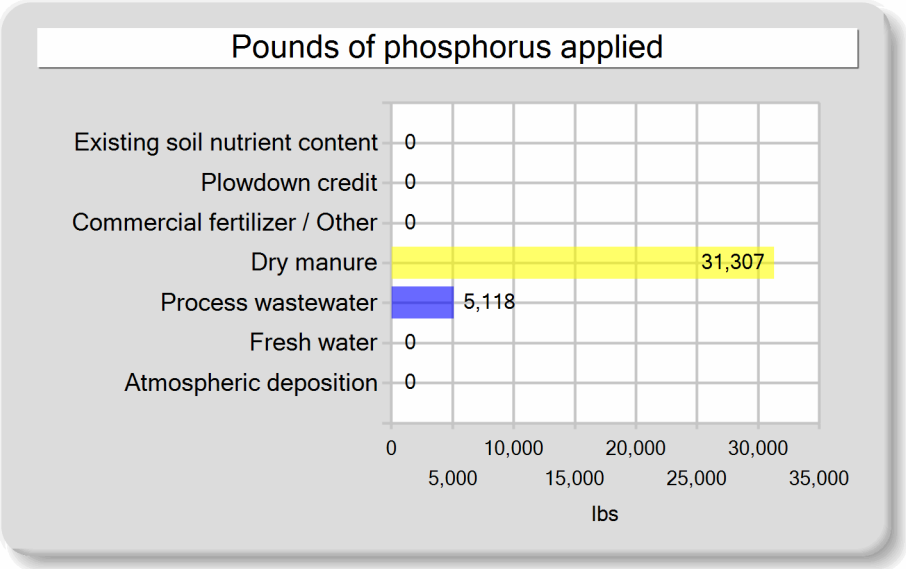
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	77,379.39	31,307.15	117,207.12	411,593.85
Process wastewater	38,790.66	5,118.36	35,243.02	185,018.66
Fresh water	17,342.94	0.00	0.00	0.00
Atmospheric deposition	3,514.00	0.00	0.00	0.00
Total nutrients applied	137,027.00	36,425.50	152,450.14	596,612.51
Anticipated crop nutrient removal	105,354.00	22,336.00	140,705.60	0.00
Actual crop nutrient removal	100,227.66	22,361.47	147,813.10	647,617.97
Nutrient balance	36,799.34	14,064.03	4,637.04	-51,005.45
Applied to removed ratio	1.37	1.63	1.03	0.92

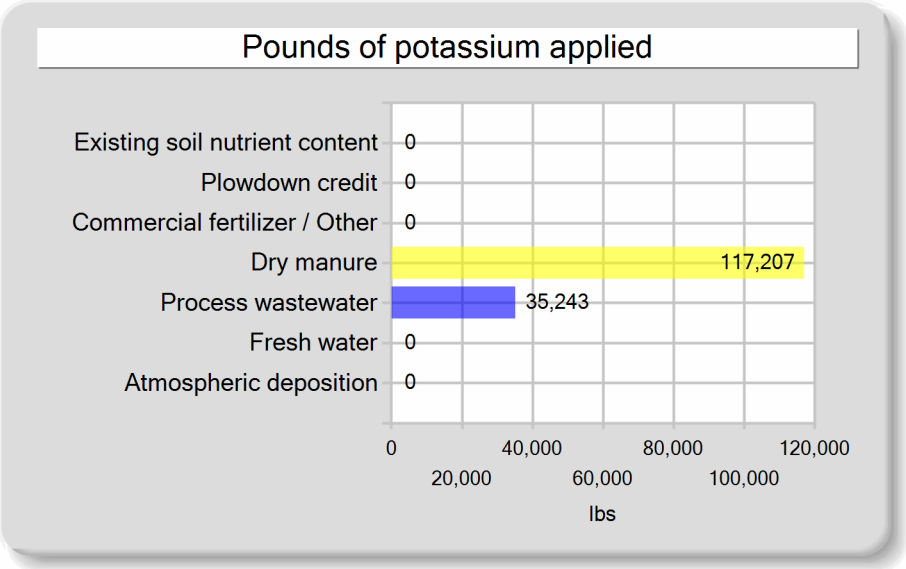
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

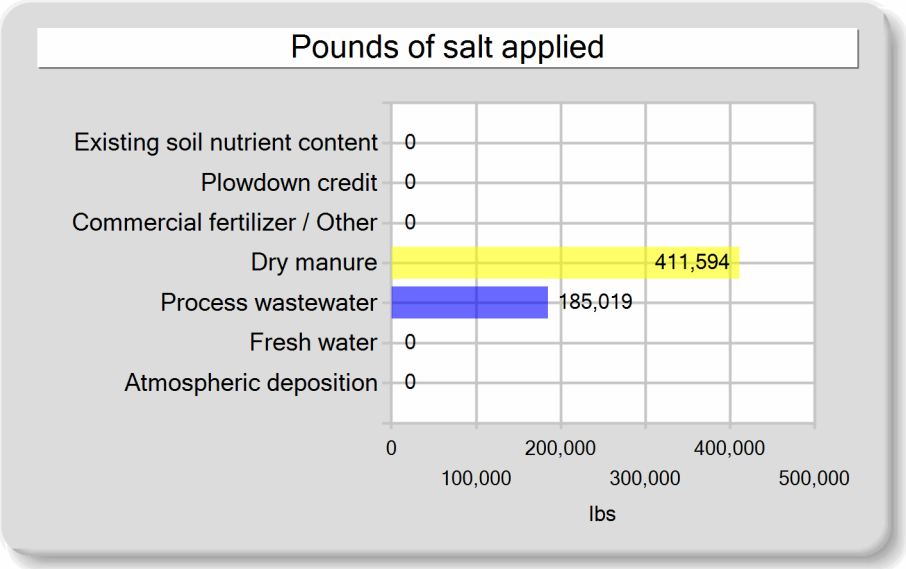


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Fields 6 & 8 show 1.42 and 1.53 respectively for corn. This is caused by the high test for N on Ag Well 6. The Ag well is being checked for unusually high test. Applications are well within agronomic rates.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
James Silva	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/24/24	6/24/24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ____James Silva_____

Name of Dairy Facility: __SB Farms_____

Facility Address: _11450 Carroll Rd._____Elk Grove_____95757__
Number and Street City Zip Code

Contact Person Name and Phone Number: __James Silva_____209-321-1318_
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _TAJ Silva Farms_____

Address of Hauling Company /Person: _12997 E. Peltier Rd._____Acampo_____95220_
Number and Street City Zip Code

Contact Person: ___James Silva_____209-321-1318_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____(please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_____SAME_____
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

__216 Hwy. 220_____Ryde_____95680_____X142-0100-X031-XXXX__
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: ____Oct 9-11, 2023_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 1.54%
P 0.487% K 2.48%

Manure: ___1000_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): ___10.8%_____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: 100 Loads hauled at approximately 10 tons per load.

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes ___X___ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 4/9/24

Hauler's Signature: _____ Date: 4/9/24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ____James Silva_____

Name of Dairy Facility: __SB Farms_____

Facility Address: _11450 Carroll Rd._____Elk Grove_____95757__
Number and Street City Zip Code

Contact Person Name and Phone Number: __James Silva_____209-321-1318_
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _TAJ Silva Farms_____

Address of Hauling Company /Person: _12997 E. Peltier Rd._____Acampo_____95220_
Number and Street City Zip Code

Contact Person: __James Silva_____209-321-1318_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____(please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_____SAME_____
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

__216 Hwy. 220_____Ryde_____95680_____X142-0100-X031-XXXX__
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: ____May 22-24, 2023_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 1.30%
P 0.351% K 2.02%

Manure: ___2000_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): ___22.4%_____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: 200 Loads hauled at approximately 10 tons per load.

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes ___X___ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 4/9/24

Hauler's Signature: _____ Date: 4/9/24

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: SB Farms
Mailing address: 22600 N. Pearl Rd.
(not facility address) Acampo, CA 95220
Phone#: (209) 321-1318

Authorized Copy Released To:
JCRK, INC.
(209)327-0992
Sampled By: Anthony Silva
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	11/9/23	1:30p	Dairy/Dom. #1	Dom. Well	11.2°C	20 min	X				
2	✓	2:00p	Dairy/Dom. #2	✓	11.2°	✓	X				
3	✓	2:30p	Ag Well Field #3	Ag Well	11.0°	✓	X				
4	✓	3:00p	Ag Well Field #6	✓	11.0°	✓	X				
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Jacquie Capteen	JCRK		11/10/23 8:00a		
2nd		DENELE ANALYTICAL, INC.	11-10-23 1:00			T3314020

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
SB Farms
12997 E. Peltier Rd
Acampo, CA 95220

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy/Dom. #1	T3314020A	Electrical Conductivity (EC)	EPA 120.1 and	0.474	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.09	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	304	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

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Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
SB Farms
12997 E. Peltier Rd
Acampo, CA 95220

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy/Dom. #2	T3314020B	Electrical Conductivity (EC)	EPA 120.1 and	0.650	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.50	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	416	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
SB Farms
12997 E. Peltier Rd
Acampo, CA 95220

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well Field #3	T3314020C	Electrical Conductivity (EC)	EPA 120.1 and	0.265	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.343	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	170	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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Compliance Analysis Report

Customer
SB Farms
12997 E. Peltier Rd
Acampo, CA 95220

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well Field #6	T3314020D	Electrical Conductivity (EC)	EPA 120.1 and	1.28	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	40.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	821	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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